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Systematic Review

Identifying indicators for measuring appropriateness of antibiotic prescribing in high-income countries: a rapid systematic review

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ABSTRACT

Background: Antibiotics are essential for treating and preventing bacterial infections, yet inappropriate use drives the development of antimicrobial resistance, posing a major global health challenge. Antimicrobial stewardship aims to optimize antibiotic use, making the definition and measurement of ‘appropriate’ prescribing critical.

Objectives: The primary objective was to identify indicators used to measure appropriateness of antibiotic prescribing in high-income countries (HICs). The secondary objective was to describe levels of inappropriate prescribing in HICs.

Methods: A rapid systematic review (PROSPERO registration no: CRD42024628584) was conducted using Embase, Medline, and Cochrane databases. Eligible peer-reviewed studies published from 2014 to January 2025 reported indicators for measuring appropriateness of antibiotic prescribing in HICs. Each article was independently reviewed for inclusion, extracted, and assessed for risk of bias by one reviewer, with 10% verified at each stage.

Results: This rapid review identified 103 unique indicators from 165 studies: 58 patient-specific/prescription-related (PSPR) indicators from 128 studies and 45 proxy indicators from 38 studies. The most frequent PSPR indicator was compliance with guidelines (95/128, 74.2%). This was followed by indicators related to appropriate duration (70/128, 54.7%) and appropriate dose (60/128, 46.9%). The most common proxy indicator was rate of prescribing by indication (22/38, 58%) followed by correct choice of antibiotic according to indication (14/38, 36.8%). Indicators were applied across diverse settings, populations, and types of infection. Among studies describing PSPR indicators, 103 of 128 gave quantifiable outcomes for inappropriateness of prescribing, which when standardized ranged from 2% to 88%.

Conclusions: This review provides a comprehensive catalogue and categorization of both PSPR and proxy indicators of antibiotic prescribing appropriateness across HICs. Recurring themes reflect shared stewardship principles: prescribing the correct antibiotic, dose, frequency, and duration, in-line with guidance, and tailored to the clinical context. These findings provide a valuable resource for those monitoring antibiotic prescribing practices. **Rachel Berry, Clin Microbiol Infect 2026;■:1**

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Introduction

Antimicrobial resistance (AMR) remains a major global health threat, responsible for an estimated 4.7 million deaths in 2021, projected to rise to 8.2 million globally by 2050 if no effective action is taken [1].

Inappropriate antibiotic use such as unnecessary prescription, dosing, or durations remains a major modifiable driver of AMR across healthcare settings [2]. Antimicrobial stewardship (AMS) programmes aim to address this issue by promoting the rational use of antibiotics, thereby preserving their effectiveness and improving patient outcomes [3].

An important step in supporting AMS efforts is the ability to define and measure what constitutes 'appropriate' or 'inappropriate' antibiotic prescribing. This requires valid and reliable indicators that can assess the quality of antibiotic use at various levels of the healthcare system [4]. Such indicators may be direct measures of appropriateness for individual patients or antibiotic prescriptions (described in this article as patient-specific/prescription-related [PSPR] indicators) or proxy indicators; these indicators estimate appropriateness based on prescribing patterns, adherence to guidelines, or other criteria, rather than clinical assessment of individual antibiotic use [5].

Several initiatives and studies have developed or applied such indicators [6–9]. However, no standardized set of indicators has been universally adopted across high-income countries (HICs), and practices vary according to setting, target population, and clinical indication.

This rapid systematic review aimed to identify the indicators used to estimate or measure the appropriateness of antibiotic prescribing in HICs (defined as Organization for Economic Co-operation and Development member countries [10]) and to classify these by setting, population, and indication. The secondary objective was to describe levels of inappropriate prescribing in HICs.

Methods

This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines [11] with adaptations for a rapid review because of the need for prompt information to support timely policy development and resource limitations. These adaptations included restricting the publication period to studies published after 2014 to ensure current relevancy, limiting dual screening and data extraction to 10% of included

articles, and excluding grey literature as well as articles not accessible through institutional subscriptions of the project group. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist is provided in the web-only Table S1, and the recommendations by Stevens et al. [12] were also followed for reporting of this rapid systematic review. The protocol was registered with PROSPERO (CRD42024628584).

Search strategy

A systematic literature search was conducted in Embase, MEDLINE, and the Cochrane Library databases in January 2025 (R.B.).

The search strategy was developed by the expert review team (R.B., L.C., M.G.C., M.G.J.dB., E.M., F.M., R.M., N.P.-S., and D.A.O.) with support and advice from UK Health Security Agency Knowledge and Evidence specialists. It targeted studies describing, developing, or evaluating indicators or metrics used to estimate the appropriateness of systemic antibiotic prescribing (treatment or prophylaxis) in any healthcare setting in HICs. Eligibility criteria are summarized in Fig. 1.

A summary of the search strategy is shown in Table 1 (for detailed search strategy see web-only Table S2).

Screening

All identified articles were imported into Rayyan and duplicates were removed. Initially, all reviewers (R.B., M.G.C., Y.K., E.M., F.M., N.P.-S., and D.A.O.) jointly screened 10 records to refine eligibility criteria and ensure consistency. Thereafter, each article was screened by one reviewer, with 10% reviewed by a second (randomly assigned by Rayyan). Discrepancies were resolved by consensus through discussion or, if agreement could not be reached, by a third reviewer whose decision was final; full text was accessed if needed. Rayyan was used to track inclusion/exclusion decisions.

Data extraction

A standardized data extraction tool was developed in Microsoft Excel. Each article was extracted by one reviewer (R.B., M.G.C., Y.K., E.M., F.M., N.P.-S., and D.A.O.), with 10% verified in duplicate (every 10th article when ordered alphabetically by author). Discrepancies were all resolved by consensus after discussion.

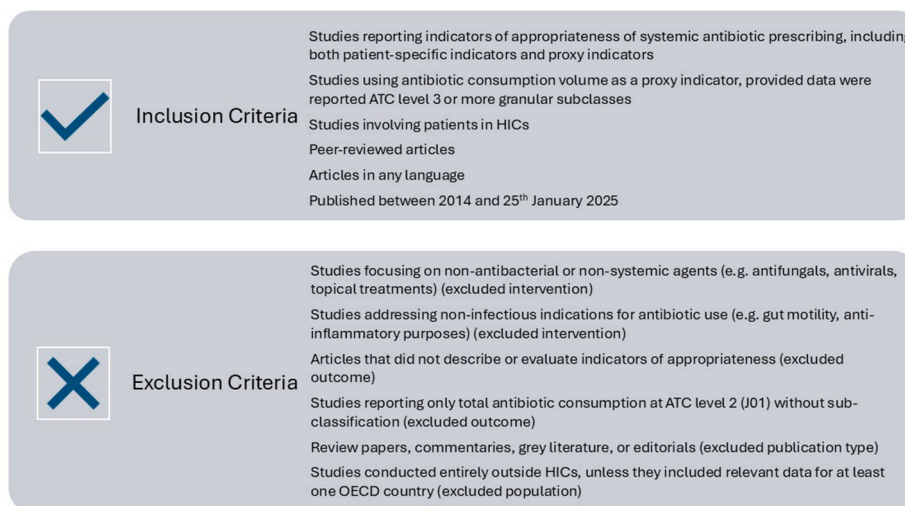


Fig. 1. Eligibility criteria.

Table 1
Summary of search strategy

Indicator	Appropriate prescribing	Antibiotic	Antibiotic prescribing or use
Criteria*	((inappropriat* or appropriat* or inadequate or	Antibio*	((antibio* or anti-bio* or anti-infective* or
Metric*	adequate or compliant or incorrect or correct or	Anti-biot*	antiinfective* or antibacterial* or anti-
Indicator*	incorrect or concordant or valid or invalid or	Antiinfective	bacterial*) adj2 (misuse or mis-use or overuse
Survey*	"evidence-based" or ineffective or effective or	Anti-infective	or over-use)).tw,kf.
Estimat*	optimal or targeted or timely or reduc* or	Antibacterial	((antibio* or anti-bio* or anti-infective* or
Audit*	decreas* or increas* or improv* or rational or	Antibacterial	antiinfective* or antibacterial* or anti-
Assess*	irrational) adj2 (prescrib* or prescription* or		bacterial*) adj1 prescri*).tw,kf.
Defin*	"use" or "usage" or consumpt* or utili*).tw,kf		

Extracted data included:

- Study characteristics (country, setting, patient population, infection type, treatment, or prophylaxis)
- Indicators used to define appropriateness of antibiotic prescribing
- How the authors identified the indicators, and whether they were validated
- Quantitative estimates of appropriateness (if available)

Risk-of-bias assessment

Risk of bias was assessed using the Mixed Methods Appraisal Tool [13]. Each article was assessed by one reviewer (R.B., M.G.C., Y.K., E.M., F.M., and B.Y.N.), with 10% reassessed by a second (the 10th article when ordered alphabetically by author, starting at the fifth article). No discrepancies were identified.

Data synthesis

Given the heterogeneity in study designs, populations, settings, and indicators, a narrative synthesis was conducted. Indicators were grouped thematically as either PSPR indicators or proxy indicators, depending on their purpose and underlying data source. They were then further subdivided by theme to identify recurring concepts or keywords from the original texts, consolidating conceptually similar indicators even if described differently. All thematic coding was done by one person with clinical expertise to ensure consistency and reliability of the analysis (R.B.), with discussion with the expert review team where needed.

PSPR indicators were grouped according to the stage of the prescribing process they evaluated (treatment decision, before prescribing, at prescribing, and after initial prescribing) and further categorized by setting (e.g. hospital and community), patient characteristics, and infection type. Proxy indicators were grouped by the outcome they measured (e.g. delayed prescribing and compliance) and the minimum data required to operationalize the indicator were identified. The thematic organization, filtering, and mapping processes were conducted using Microsoft Excel.

To quantify levels of inappropriate prescribing, we analysed those studies that reported PSPR indicators and provided quantitative measures of appropriateness. Definitions of appropriateness varied across studies, with some reporting binary categories 'appropriate' or 'inappropriate', whereas others included multiple categories such as 'appropriate', 'inappropriate', 'unnecessary', or 'unclassifiable'. Data were therefore standardized when needed to reflect the proportion of inappropriate prescribing. For example:

- When only the percentage of appropriate prescriptions was reported, the remaining proportion was assumed to be inappropriate.
- For pre-post intervention studies, only baseline data were used.

- For studies reporting inappropriateness of multiple years or infection types, a nonweighted average was calculated.

Subgroup analyses were performed by setting, as either community (including primary care, long term, and dental settings) and hospital care.

Results

The searches returned 2046 records. After removing duplicates, 1495 unique papers were screened on title and abstract. Of these, 1235 were excluded, leaving 260 for full-text review. Ten articles could not be retrieved, and a further 85 were excluded because they met exclusion criteria, leaving 165 eligible studies (see Fig. 2 and web-only Table S3).

A summary of article characteristics is provided in Table 2 [5,6,9, 14–50, 51–100, 101–175]. For the full extraction table see web-only Table S3.

Most of the articles were observational studies such as point prevalence audits (98/165, 59.4%), followed by Quality Improvement studies including prospective audit and feedback (26/165, 15.8%), consensus-building studies (18/165, 10.9%), and interventional studies (14/165, 8.5%). The remainder were mixed method studies, study protocols, and randomized trials (3/165, 1.8% for each).

Most articles focused on the hospital or inpatient settings (103/165, 62.4%) and did not specify the age group of the patient population (94/165, 56.9%). The majority addressed treatment (87/165, 52.7%) rather than prophylaxis and 58.8% (97/165) did not specify a particular clinical indication.

Risk-of-bias assessment

Overall, studies were considered to be of moderate to good quality, with 103 of the included studies (62.4%) fulfilled 100% of the relevant quality criteria. Eight studies fulfilled less than 50 (4.8%) and 5 studies were excluded from risk-of-bias assessment because they did not report methods or results (see web-only Table S4 and Fig. S5).

Indicators

PSPR indicators

A total of 127 studies described PSPR indicators of appropriateness (127/165, 77%), and one described both PSPR and proxy indicators (1/165, 0.7%). In total, 58 distinct PSPR indicators were identified as measures of appropriate antibiotic prescribing in HICs (see Table 3 and web-only Table S6).

The most frequently reported PSPR indicators of appropriate antibiotic prescribing were compliance with local, national, or regional guidelines (95/128, 74.2%), appropriate duration of therapy (70/128, 54.7%), appropriate dose (60/128, 46.9%), and

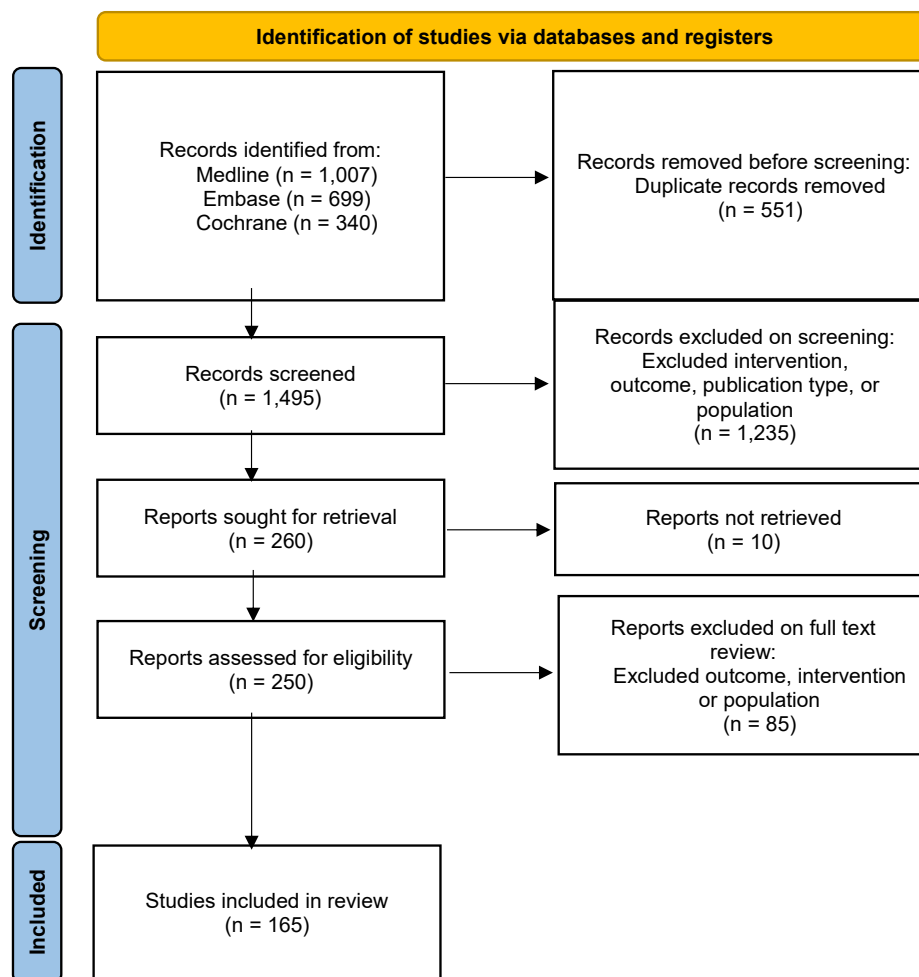


Fig. 2. Flowchart of study selection.

appropriate choice of antibiotic for the documented indication (57/128, 44.5%). These indicators reflect core AMS principles.

Most articles (92/128, 71.8%) employed composite definitions of appropriate prescribing, where multiple indicators were required to be met for a prescription to be classified as appropriate [6,9,14,16–19,24,25,30,31,36,37,41–47,49,50,52–54,56,59,62,66–72,77–83,85–90,93,95,103–107,110–117,120,124–128,130,133,134,137,139,140,142,144,149,151–153,155,157,159,160,163,164,166,168,169,171,174]. These definitions incorporated between 2 and 16 individual indicators.

When analysed by setting, hospitals demonstrated the broadest application of PSPR indicators, utilizing 57 of the 58 identified. In contrast, primary care, outpatient, and ambulatory care settings employed a more limited set (34/58 PSPR indicators). In long-term or residential care settings, 13 indicators were reported, and only 8 were reported in dental settings.

By patient age group, the widest range of PSPR indicators was reported in studies involving adults (50/58 indicators). The remaining eight indicators appeared only in studies that did not specify any age group, suggesting potential applicability to adult populations. A total of 27 indicators were described in paediatrics, with a narrower range used in neonatal patients, with only 9 indicators reported.

By infection type, fewer PSPR indicators were reported. Urinary tract infections were associated with 23 of 58 indicators and respiratory tract infections with 22 of 58 indicators. Dental

prophylaxis and treatment, prophylaxis of surgery or procedures, and sepsis or bacteraemia were each associated with 17 indicators; a narrower range was described for other identified infections.

Proxy indicators

Thirty-seven articles described proxy indicators for assessing or estimating appropriateness (37/165, 22.3%) and one described both PSPR and proxy indicators (1/165, 0.7%).

A total of 45 proxy indicators were identified to assess appropriateness of antibiotic prescribing in HICs (Table 4 and web-only Table S7).

The most frequently reported proxy indicators of appropriate prescribing were the rate of prescribing by indication (22/38, 57.9%), the proportion of patients receiving first-line treatment according to indication (14/38, 36.8%), and seasonality of total antibiotic use <20% (8/38, 21.1%).

When analysed by setting, primary care demonstrated the broadest application of proxy indicators, with 37 of the 45 identified. Long-term care settings employed a narrower subset of 11 proxy indicators. In contrast, hospital, dental, and community pharmacy settings were associated with only a limited number of proxy indicators (5, 7, and 2, respectively).

By age group, the widest range of proxy indicators was reported in studies involving adults, with 19 of the 45 identified proxy indicators. However, 40 of the 45 were either reported for adults or

Table 2
Summary of characteristics of included studies

Characteristics	Number of articles (%) Total = 165	References
Country		
United States	33 (20%)	[14–46]
United Kingdom	17 (10%)	[9,47–62]
France	15 (9.1%)	[5, 63–76]
Australia	14 (8.5%)	[77–90]
Netherlands	13 (7.8%)	[91–103]
Spain	11 (6.7%)	[6,104–113]
Multicountry	10 (6%)	[114–123]
Canada	8 (4.8%)	[124–131]
Italy	6 (3.6%)	[132–137]
Japan	6 (3.6%)	[138–143]
Germany	5 (3%)	[144–148]
Korea	4 (2.4%)	[149–152]
Switzerland	4 (2.4%)	[153–156]
Belgium	3 (1.8%)	[157–159]
Greece	3 (1.8%)	[160–162]
Ireland	3 (1.8%)	[163–165]
Norway	3 (1.8%)	[166–168]
Türkiye	2 (1.2%)	[169,170]
Austria	1 (0.6%)	[171]
Denmark	1 (0.6%)	[172]
Estonia	1 (0.6%)	[173]
New Zealand	1 (0.6%)	[174]
Sweden	1 (0.6%)	[175]
Healthcare setting		
Hospital, including private settings	103 (62.4%)	[6,9,14,15,18,22,24,30], [31,33–36,40,43,45–59], [62,64–68,71,72,77–80], [82,83,85–99], [104–107,111–113,115,117,120], [122,124–128,130], [132–134,136,137,139,140], [142,144,145,148,153,155], [157–162,164,166–169,171,174,175]
Primary, community setting, or ambulatory care	44 (26.7%)	[5,16,17,19,20,23,26,28,29], [32,37–39,44,60,61,69,73,76], [81,84,100–102,108–110], [118,119,121,123,129,131,135,138], [141,147,152,154,156], [165,170,172,173]
Long-term care or residential care setting	6 (3.6%)	[21,27,63,70,75,163]
Multiple settings	6 (3.6%)	[103,114,116,149–151]
Dental	5 (3%)	[25,41,42,74,146]
Community pharmacy	1 (0.6%)	[143]
Population		
Not specified	94 (56.9%)	[5,6,9,14,17–19,22–24,29], [31–33,35–37,39,40], [45,48–50,52,53,55,56,58], [66–68,71–73,77–84,88–93,95,96], [100–103,106,108,110], [112,114–116,118,121,124–127], [134,136,139–149], [151,153–155,157,158,160], [161,164,166,169,171,173,175]
Adults	41 (24.8%)	[16,21,25–27,30,41–44,47], [54,57,59,63,70,74,75,86], [94,97–99,104,105,107,111,117,120], [122,128–130,133,135], [152,156,159,163,170,174]
Children	16 (9.7%)	[15,20,34,46,51,61,62,76,87,113,119,137,138,162,167,168]
Adults and children	11 (6.7%)	[28,38,60,69,109,123,131,132,150,165,172]
Older adults >75 y	2 (1.2%)	[64,65]
Neonates <28 d	1 (0.6%)	[85]
Treatment or prophylaxis		
Treatment	87 (52.7%)	[5,6,9,15–17,19–24,26,28–30], [32,34,37–39,43,44,49,50], [52,56,57,59–61,63–66,70–76], [81,84,85,88,93,94,96,98–105], [107–112,117,121,122,125–131], [133,137,142,143,147,149,154], [156,158,159,165,166,170,172]
Both	59 (35.8%)	[14,25,27,33,36,40,45–47], [51,53,58,62,67,69,77–80], [83,86,87,90,91,95,97], [106,113–116,120,123,124,135], [136,138–140,144–146], [148,150–152,155,157], [161–164,167–169,171,173–175]
Prophylaxis	10 (6%)	[31,35,41,42,48,54,55,82,132,160]
Not specified	9 (5.5%)	[18,68,89,92,118,119,134,141,153]
Clinical indication		
Not specified	97 (58.8%)	[5,6,9,14,15,18,24,30,33,34,43,45,46,49], [51,52,56–58,63–69,71–73], [75–81,83,86,87,91–95,97,99], [101,103,104,106–108,111–113,115], [116,118–120,122,124–128,130], [134–141,143–145,148,150–153,155], [157–159,161,163,164,167–169], [171,173–175]
Respiratory tract infections	18 (10.9%)	[16,20,23,29,38,39,44,61,84,110,117,121,123,133,165,166,170,172]
Various	11 (6.7%)	[28,32,40,53,60,100,102,109,129,131,147]
	9 (5.5%)	[25,36,41,42,50,62,74,90,146]

(continued on next page)

Table 2 (continued)

Characteristics	Number of articles (%) Total = 165	References
Dental or oral infections or prophylaxis		
Urinary tract infection	8 (5%)	[22,26,27,70,96,149,156,162]
Prophylaxis of surgical or procedural infection	7 (4.2%)	[31,35,48,54,55,82,160]
Sepsis or bacteraemia	5 (3%)	[85,89,98,105,142]
Urinary tract infections and respiratory tract infections	4 (2.4%)	[17,21,114,154]
Urinary tract Infection and skin and soft tissue infections	1 (0.6%)	[19]
Healthcare-associated infections	1 (0.6%)	[132]
Acute pancreatitis	1 (0.6%)	[47]
Skin and soft tissue infections or postsurgery infections	1 (0.6%)	[88]
Suspected or confirmed intra-abdominal infection	1 (0.6%)	[59]
Chlamydia trachomatis infection	1 (0.6%)	[37]
Specific criteria		
Patients prescribed specific antibiotics	18 (10.9%)	[24,43,49,58,59,66–69,86,104,107,112,125,128,130,143,159]
Patients in the emergency department	5 (3%)	[45,79,93,94,111]
Patients on critical or intensive care	4 (2.4%)	[80,92,120,127]
Veterans	4 (2.4%)	[16,41–43]
Patients in the dental department	3 (1.8%)	[36,50,62]
Medical patients	3 (1.8%)	[56,57,134]
Oncology or patients with cancer	2 (1.2%)	[22,54]
Patients on outpatient parenteral antibiotic therapy (OPAT)	2 (1.2%)	[81,109]
Hospitalized older adults	2 (1.2%)	[64,65]
Patients prescribed intravenous (i.v.) antibiotics	2 (1.2%)	[97,169]
Patients prescribed oral antibiotics	2 (1.2%)	[14,15]
Haemodialysis patients	1 (0.6%)	[30]
Method used for indicator identification		
Literature review	46 (27.9%)	[5,6,15,22,24,27,29,32,34,35,38–40,45], [54,58,67–69,71,72,75,76,78], [93,95,97,100,106,108,109], [113,117,120,129,139,140,142], [144,149,152,153,159,162,169,172]
Using treatment or diagnostic guidelines	35 (21.2%)	[16,19–21,23,25,26,30,31,36,37,41,42,44,48,50,55,59,62,66,74,84,99,105,107,110,121,125,128,133,137,155,160,166,170]
Not specified	30 (18.2%)	[14,17,18,33,47,57,87,91,103,104,111,112,114,115,119,124,130,132,134,136,138,143,145,148,161,164,167,168,171,175]
Pre-existing audit tool	16 (9.7%)	[46,52,56,77,79–82,85,86,88–90,126,158,174]
Delphi consensus	15 (9.1%)	[9,63,65,73,92,94,96,98,122,127,131,146,150,157,163]
Pre-existing indicator	12 (7.3%)	[51,61,101,118,123,135,141,147,154,156,165,173]
Expert group	11 (6.7%)	[28,43,49,53,60,64,70,83,102,116,151]
Validation status of the indicators		
Yes – in this study	62 (37.6%)	[5,9,17,20–24,26,29,31,32,34,35,37,39], [42–44,47–50,53,56,57,63,65–67,70], [71,73,76,78,83,93,96,98,99], [101–104,106,108,109,116,117], [122,125,128,140,144–146], [151,152,154,160,166,171]
Yes – in previous study	47 (28.5%)	[6,27,38,40,45,46,58,61,68,69], [72–74,77,79,80,82,85,86,88,89,95,97], [110,118,120,121,123,126,129,132,133,135], [136,139,147,153,155,156,158,159], [161,165,169,173–175]
No	32 (19.4%)	[15,18,30,33,41,51,52,54,59,60,62,90,92,94,100,105,113,114,124,127,130,131,134,137,138,143,148,150,157,162–164,167]
Not specified	24 (14.5%)	[14,16,19,25,28,36,55,60,64,81,84,87,91,107,111,112,115,119,141,142,149,168,170,172]
Indicator type described		
Patient-specific/prescription-related (PSPR) indicators	127 (77%)	[6,9,14–19,21–27,30,31,33], [35–37,41–50,52–59,62,64–72], [77–83,85–99,103–108,110–117], [120–122,124–128,130,132–134], [136–140,142,144,145,149–153], [155,157–164,166–171,174,175]
Proxy indicators	37 (22.3%)	[5,20,28,29,32,34,38–40,51,60,61,63,73–76,84,100–102,118,119,123,129,135,141,143,146–148,154,156,165,172,173]
Both	1 (0.7%)	[109]

Table 3
Patient-specific/prescription-related (PSPR) indicators used to assess appropriateness of antibiotic prescribing in HICs

Patient-specific/prescription-related (PSPR) indicators of appropriateness	Number of studies (n = 128)	Settings				Ages				Treatment or prophylaxis				Specified infections				Specified patient groups													
		Hospital	Primary, community or ambulatory care	Long-term or residential care	Dental	Not specified	Adults	Children	Neonates	Not specified	Treatment	Prophylaxis	Both	Not specified	Respiratory tract infection	Urinary tract infection	Dental prophylaxis or infections	Prophylaxis of surgery or procedure	Skin or soft tissue infections	Sepsis or bacteraemia	Other	Specific antibiotics	Intensive care	Veterans	Emergency department	Medical patients	Oncology patients	Outpatient parenteral antibiotic therapy (OPAT)	Intravenous antibiotics	Other	
Treatment decision																															
Is decision compliant with local, regional, or national guidelines?	95	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
Is decision in-line with expert opinion?	36	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x		x		x	
Is decision in-line with evidence-base or scientific information?	6	x		x		x	x				x		x	x		x						x	x								
Is decision in-line with decision support tools?	1	x				x							x	x																	
Before prescribing - need for antibiotic																															
Are antibiotics indicated?	21	x	x	x	x	x	x	x			x	x		x	x	x	x	x			x		x		x			x	x		
Has the patient got a confirmed or suspected bacterial infection?	17	x	x	x		x	x				x			x	x	x		x				x	x			x				x	
Has the patient got clinical signs of an infection?	13	x	x			x	x				x			x	x	x			x			x		x	x						
Have relevant diagnostic criteria been met?	13	x	x	x		x	x	x			x	x		x	x	x			x							x					
Have diagnostic blood cultures been taken prior to antibiotic prescribing?	12	x	x			x	x	x			x	x		x	x					x		x							x	x	
Have diagnostic cultures from suspected site of infections been taken prior to antibiotic prescribing?	12	x	x			x	x	x			x			x	x					x									x		
Is the patient at risk for an infection?	6	x	x		x	x	x				x	x		x	x	x		x						x							
Have relevant diagnostic tests been carried out?	3		x			x	x				x				x									x							
Is the patient classified as palliative, meaning that antibiotics may not be appropriate?	1	x					x				x								x												
Have relevant severity criteria been met?	1	x					x				x			x																	
At prescribing																															
Is the duration correct?	70	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
Is the dose correct?	60	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x		x		x	x	x	x	
Is the choice correct?	57	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
Is the route correct?	33	x	x			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x				x	x			
Is the frequency correct?	25	x	x			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x			x	x	x			
Is the timing of administration correct?	15	x	x			x	x	x			x	x		x	x			x		x		x			x						
Is the medicinal form correct?	2	x					x	x			x						x				x	x									
Have the following factors been considered when choosing the regimen																															
The indication being treated?	28	x	x	x		x	x	x			x	x	x	x	x	x	x	x				x		x	x		x				
The activity of the antibiotic against the known or suspected organism?	26	x	x	x	x	x	x	x			x	x		x	x	x		x	x			x	x		x						
The patient's allergy status?	25	x	x			x	x	x	x	x	x	x	x	x		x	x		x			x	x	x	x			x			
Whether the spectrum of the antibiotic is too broad or too narrow?	24	x	x			x	x	x	x	x	x	x		x	x		x	x	x	x		x	x		x			x	x		
Whether there is any duplication, overlap or redundancy in the regime?	19	x	x			x	x	x			x	x		x	x							x	x	x	x			x	x		
The patient's renal function?	15	x	x			x	x	x			x			x	x	x	x					x			x					x	
The patient's previous antibiotic use?	6	x	x			x	x				x				x	x						x		x	x						
Whether the i.v. or oral route is appropriate based on specified criteria?	6	x				x	x	x			x			x						x	x	x									

(continued on next page)

Table 3 (continued)

Patient-specific/prescription-related (PSPR) indicators of appropriateness	Number of studies (n = 128)	Settings				Ages				Treatment or prophylaxis				Specified infections								Specified patient groups									
		Hospital	Primary, community or ambulatory care	Long-term or residential care	Dental	Not specified	Adults	Children	Neonates	Not specified	Treatment	Prophylaxis	Both	Not specified	Respiratory tract infection	Urinary tract infection	Dental prophylaxis or infections	Prophylaxis of surgery or procedure	Skin or soft tissue infections	Sepsis or bacteraemia	Other	Specific antibiotics	Intensive care	Veterans	Emergency department	Medical patients	Oncology patients	Outpatient parenteral antibiotic therapy (OPAT)	Intravenous antibiotics	Other	
The patient's severity criteria?	5	x	x			x	x			x			x	x						x		x									
Whether combination or co-prescribing is needed?	5	x				x	x	x		x			x	x								x									
The patient's contraindications?	5	x				x	x			x			x	x								x	x		x						
Whether an inappropriate combination has been prescribed?	4	x	x				x	x		x			x	x		x															
The patient's previous culture results, including colonisation?	4	x	x			x	x			x			x	x				x				x		x	x						
The patient's gender?	4	x	x			x	x			x				x			x														
The patient's co-morbidities?	3	x	x	x			x	x		x			x	x	x																
The patient's weight?	3	x				x	x	x		x			x	x												x					
Interactions with the patient's other medicines or feeds?	3	x				x	x		x	x			x	x								x		x							
Whether the patient is pregnant?	3	x	x			x	x			x				x			x					x									
The patient's risk of adverse events?	2	x				x	x			x			x	x	x							x									
Whether the antibiotic will penetrate to the target site?	2	x				x			x	x				x									x				x				
The patient's risk factors for resistant bacteria?	2	x	x			x				x			x	x								x									
The mode of action of the antibiotic?	1	x					x						x	x								x									
Whether the patient has a catheter?	1	x				x				x							x														
The patient's immune status including immunosuppressive medicines?	1	x					x			x				x																	
The patient's travel history?	1	x				x							x	x																	
The practicality of the antibiotic?	1	x				x							x	x											x						
After initial prescribing																															
Have cultures been reviewed and treatment adapted accordingly?	35	x	x	x		x	x	x	x	x	x	x	x	x		x	x	x	x	x		x			x	x				x	
Has there been appropriate documentation in the clinical record?	29	x	x			x	x	x		x	x		x	x				x				x	x		x	x		x		x	
Has intravenous treatment been reviewed and switched to oral treatment (IVOS) appropriately?	16	x	x			x	x	x		x	x		x	x	x	x						x	x								
Has appropriate Therapeutic Drug Monitoring taken place?	9	x				x	x			x	x		x	x					x				x								
Has the treatment been de-escalated?	6	x				x	x	x		x	x		x	x		x							x								
Has the treatment been stopped if there is no evidence of infection?	5	x				x	x				x		x	x										x							
Has the treatment been reviewed or reassessed?	5	x	x			x	x			x	x		x	x				x												x	
Has the treatment been effective?	2	x				x							x	x											x						
Has the treatment been monitored?	2	x				x	x			x			x	x								x									
Is the patient taking it?	1	x				x							x	x																	
Has the patient been educated on how to take it?	1	x					x			x				x											x						

HIC, high-income country.

Table 4
Proxy indicators used to assess appropriateness of antibiotic prescribing in HICs

Proxy indicators of appropriateness	Number of studies (n = 38)	Range of target (where specified)	Setting					Age			Infection							Minimum data required to operationalize indicator				
			Hospital	Primary, community, or ambulatory care	Long-term or residential care	Dental	Community pharmacy	Not specified	Adults	Children	Not specified	Respiratory tract infection	Urinary tract infection	Skin or soft tissue infections	Dental infections	Other	Consumption data	Patient-linked prescription details (choice, dose, duration)	Patient and prescription-linked diagnosis/infection information	Additional patient information		
Consumption indicators																						
Fluoroquinolones	6			x					x	x		x						x				
Penicillins plus beta-lactamase inhibitor	5			x					x	x		x						x				
Third- and fourth-generation cephalosporin	5			x				x	x	x		x						x				
Broad/narrow ratio	5			x		x			x	x	x	x						x				
First-line/second-line ratio	4	>1		x		x			x	x	x	x						x				
All penicillins	3			x					x			x						x				
Beta-lactamase sensitive penicillins	3			x					x			x						x				
Macrolides, lincosamides, and streptogramins	3			x					x			x						x				
Cephalosporins	3			x					x			x						x				
Amoxicillin/amoxicillin-clavulanate ratio	2	<10		x			x		x	x		x				x		x				
Macrolides	2			x					x			x						x				
Amoxicillin-clavulanate + Macrolides + Quinolones	2			x					x			x						x				
Penicillin	1						x		x							x		x				
Amoxicillin	1			x							x	x						x				
Amoxicillin-clavulanate	1			x					x			x						x				
Azithromycin	1			x							x	x						x				
Clindamycin	1						x		x							x		x				
Moxifloxacin	1			x					x			x						x				
Amoxicillin + cefalexin	1			x							x	x						x				
Pristinamycin + macrolides	1			x					x			x						x				
Carbapenems and piperacillin-tazobactam	1		x								x	x						x				
Drugs that make up 90% of antibiotic volume (DU90)	1			x					x			x						x				
Antibiotic diversity	1		x								x	x						x				
% 'Access' category	1			x					x			x						x				
Route																						
Intravenous antibiotics	2		x						x			x						x				
Duration																						
Duration greater than specified number of days	5	<10%		x		x		x	x										x			
Days of therapy	2		x					x					x		x			x	x			
Mean duration	1			x							x	x		x		x		x	x			
Seasonality																						
Seasonality of total antibiotic use	8	<20%		x		x			x	x	x	x						x				
Seasonality of quinolone use	6	<10%		x		x			x	x		x						x				
Seasonality of amoxicillin-clavulanate use	1					x				x		x						x				
Patient-specific indicators																						
Need for antibiotic																						
Patients prescribed an antibiotic based on the specified indication	22	Based on indication	x	x			x		x	x	x		x		x		x		x			
Patients having appropriate diagnostic tests before prescription of antibiotic	3			x					x		x		x						x	x		
Patients meeting required diagnostic criteria before prescription of antibiotic	2	90–100%		x					x				x						x	x		
Choice																						
Patients prescribed the first-line choice for the specified indication	14	80–100% 90–100% 50–100%		x		x			x	x	x	x	x		x		x		x	x		
Patients prescribed a fluoroquinolone for a specified indication	5	<10%		x					x	x	x		x		x				x	x		
Patients coprescribed a fluoroquinolone and steroid	5	<5%		x		x			x	x	x	x							x			
Patients prescribed antibiotic and non-steroidal anti-inflammatory drug (NSAID) on the same day	5	<5%		x		x			x	x	x	x							x			
Patients prescribed a fluoroquinolone within 6 mo of another prescription for a fluoroquinolone	4	<10%		x		x			x		x		x						x	x		
Patients prescribed an antibiotic that is not indicated	3	<0.5%		x		x			x		x		x						x			
Patients prescribed an inappropriate antibiotic choice for the indication	2	<1% <20%		x				x		x			x				x		x			
Patients prescribed a rarely used antibiotic choice for the indication	1	<5%					x				x					x			x			
Patient factors																						
Patients without known penicillin allergy prescribed a pen-allergic treatment option	1	<10%		x					x				x						x	x		
Pregnant patients prescribed doxycycline	1			x					x			x							x			
Male patients receiving first-generation fluoroquinolones, nitrofurantoin, or Fosfomycin	1			x					x			x							x	x		

HIC, high-income country.

not age-specified, suggesting broad applicability to adult populations. In children, 15 proxy indicators were described, 6 of which were unique to this group.

When classified by infection type, a narrower scope was observed. Respiratory tract infections were associated with eight proxy indicators, dental infections with seven, urinary tract infections with four, skin and soft tissue infections with three, and other infections with two.

Levels of inappropriateness of antibiotic prescribing

A total of 103 articles describing PSPR indicators provided quantifiable information on inappropriateness of prescribing (103/165, 62.4%) [6,14–17,19,21–26,30,31,35–37,41–50,52–59,62,66–72,77–83,85–90,93,95,104–108,110,111,113,116,117,120,121,124–126,128,130,132–134,136–140,144,149,151,152,155,158–162,164,166–171,174,175].

Reported levels of inappropriate prescribing ranged from 2.2% to 88%. Hospital and multisetting studies spanned this entire range, although most (53/82, 64.6%) were below 40%, whereas community-based studies (primary care, dental, and long-term care facilities) reported inappropriateness between 11% and 85% (see web-only Fig. S8). No trend towards improvement or decline in percentage of inappropriate antibiotic treatment over the years was identified in this review (see web-only Fig. S9).

Discussion

Indicators

This review identified 103 unique indicators—58 PSPR and 45 proxy indicators—used to assess the appropriateness of systemic antibiotic prescribing across 22 HICs. These indicators varied widely in their purpose, data requirements, and clinical contexts, with most applied in hospital settings and treatment-related decisions.

The breadth and diversity of indicators underscore the complexity of defining appropriateness of antibiotic prescribing. Many were composite, combining agent choice, dose, route, duration, and clinical justification, highlighting that appropriateness is multidimensional and rarely captured by a single metric. This mirrors the complexity of stewardship practice, where prescribing decisions must balance patient, pathogen, and clinical context [176]. However, the lack of standardization in indicator definition and application limits comparability across studies and undermines benchmarking. National initiatives such as Australia's national antimicrobial prescribing surveys demonstrate the value of defined, harmonized indicators for repeated measurement and feedback [8]. Proxy indicators, although less granular, offer scalable alternatives where patient-level data are not available.

Variation by setting, population, and infection

PSPR indicators were predominantly applied in hospitals, where audits, detailed clinical data, stewardship teams, and integrated clinical information systems facilitate access to detailed patient-level data [177]. In contrast, proxy indicators such as prescribing volume, guideline adherence, or seasonal variation were more common in primary care, where clinical justification is harder to extract, audits occur less frequently, and electronic records are less integrated within prescribing systems [73]. The relatively small number of primary care studies may also reflect lower inclusion of these settings in national stewardship programmes such as the Point Prevalence Survey. Although

digitalization is advancing in hospitals, similar progress in primary care could enable access to richer patient-level prescribing data across settings in HICs. This could enable more nuanced assessments of appropriateness, but further research is needed to explore acceptability and resource needs for such systems.

Few studies explicitly reported patient age, restricting applicability to paediatrics or older adults, where prescribing recommendations differ substantially. Similarly, infection-specific indicators were underused, despite their potential to support more targeted feedback for high-impact conditions such as respiratory or urinary tract infections.

Despite heterogeneity, recurring themes emerged: indicators frequently assessed whether antibiotics were prescribed when indicated, whether the choice, dose, frequency, and duration were appropriate, and whether prescribing followed clinical guidance. These recurring themes reflect core principles of AMS and provide a foundation for future harmonization.

Levels of inappropriateness of antibiotic prescribing

Reported rates of inappropriate prescribing for PSPR indicators varied widely, in some cases even exceeding 80%, and showed no trend towards improvement or decline over the years. However, estimates were highly dependent of study design and definition. Proxy indicators often inferred inappropriateness from broad-spectrum use or nonadherence to guidelines, whereas PSPR indicators relied on detailed audits but differed in how they defined inappropriateness, including whether composite measures were used. This variability highlights the urgent need for transparent, standardized definitions, and validated indicators, to allow comparison and benchmarking, and accurately estimate of the level of inappropriate antibiotic prescribing to guide future stewardship interventions.

Limitations

This rapid review prioritized timeliness over comprehensiveness. Searches were limited to selected databases and to terms appearing in titles or abstracts. To ensure current relevance only peer-reviewed publications since 2014 were included and grey literature was excluded, along with a small number of articles ($n = 10$) that could not be retrieved. Searches were also not repeated after January 2025. The included studies were heterogeneous in design, populations, and outcome definitions, which limited comparability. The review excluded studies that reported only total antibiotic consumption at the Anatomical Therapeutic Chemical (ATC) level 2 category (J01-antimicrobials for systemic use) without further subclassification, as well as studies that did not specify indicators explicitly. Although some indicators may have been missed because of these limitations, the review likely achieved broad coverage of those currently applied.

Moreover, our focus was on HICs and prescribing indicators, not diagnosis. This means that important preprescription determinants of appropriateness—such as diagnostic accuracy or indication validity—are not fully addressed. Because of variation in reporting, assumptions were occasionally required to classify indicators, which may have introduced subjectivity. Similarly, limitations existed in how inappropriateness was standardized. These decisions were required to interpret heterogeneous reporting but may have affected findings. Nonetheless, because the aim was to catalogue indicators and describe inappropriate prescribing rather than synthesize outcomes, these limitations are unlikely to undermine the main findings.

Implications and future work

This review provides a catalogue of the specific indicators used in HICs and encompass both PSPR and proxy indicators. To our knowledge, this combination has not previously been presented within a single paper. This is a useful, structured resource for health authorities, AMS teams, and researchers aiming to monitor and improve antibiotic prescribing. It highlights the need for a core set of validated, standardized indicators, adaptable across settings and infection types. Proxy indicators can facilitate large-scale surveillance where detailed data are lacking, whereas PSPR indicators remain essential for in-depth audits and quality improvement. Future research should explore how indicators perform across diverse populations, how they link to patient outcomes, and how they can be integrated into electronic health record systems to support real-time feedback and benchmarking. Given the large number and diversity of indicators, this catalogue can serve as a starting point to identify which indicators are most valid, feasible, and relevant for monitoring and improving antibiotic prescribing. Prioritization of high-quality indicators will be essential to focus research and stewardship efforts, facilitate standardization, and support comparability across settings. Additional work is required to assess the quality, validity, and usability of these indicators, and to determine association with reduction in AMR.

Beyond research, the findings from this review have the potential to support policy development. Having standardized methods of assessing the appropriateness of antibiotic prescribing, based on validated and high-quality indicators will be important at both national and international levels. Achieving this will require collaboration among clinicians, researchers, policymakers, public health authorities, and professional societies, and international cooperation will be essential to ensure comparability across health systems and to facilitate shared learning.

Conclusion

This review provides a comprehensive catalogue and categorization of both PSPR and proxy indicators used to assess appropriateness of antibiotic prescribing across HICs. Despite substantial heterogeneity, recurring themes reflect shared stewardship principles: prescribing the right antibiotic, at the right dose and duration, in-line with clinical guidance. Although indicators varied in granularity and application, collectively they offer a foundation for future harmonization.

Developing a core set of validated, standardized indicators, integrated into digital health systems, will be critical to reduce variability in stewardship reporting, support benchmarking, and ultimately optimize antibiotic use across healthcare settings.

CRediT authorship contribution statement

Rachel Berry: Designed the study protocol including search terms and inclusion/exclusion criteria, conducted the searches, screened the studies and extracted the data, conducted the risk-of-bias assessment, synthesized the results, and wrote the original draft manuscript. Lucy Catteau: Designed the study protocol including search terms and inclusion/exclusion criteria and provided input to draft manuscript. Martine G. Caris: Designed the study protocol including search terms and inclusion/exclusion criteria, screened the studies and extracted the data, and conducted the risk-of-bias assessment. Markus G.J. de Boer: Designed the study protocol including search terms and inclusion/exclusion criteria. Youssa Kherabi: Screened the studies and extracted the data and conducted the risk-of-bias assessment. Eugenia Magrini: Designed the study protocol including search terms and inclusion/

exclusion criteria, screened the studies and extracted the data, and conducted the risk-of-bias assessment. Filippo Mediolì: Designed the study protocol including search terms and inclusion/exclusion criteria, screened the studies and extracted the data, and conducted the risk-of-bias assessment. Rita Murri: Designed the study protocol including search terms and inclusion/exclusion criteria. Bee Y. Ng: Conducted the risk-of-bias assessment. Nathan Peiffer-Smadja: Designed the study protocol including search terms and inclusion/exclusion criteria and screened the studies and extracted the data. Diane Ashiru-Oredope: Conceptualized the study, designed the study protocol including search terms and inclusion/exclusion criteria, screened the studies and extracted the data, and provided input to draft manuscript. All authors edited and reviewed the manuscript.

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Potential conflict of interest

The authors declare that they have no conflicts of interest.

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During the preparation of this work the authors used Microsoft Copilot to translate articles that were not in English, and to improve readability and language of the text. The tool was not used to generate original scientific content, conduct data analysis, or draw conclusions. After using this tool, the authors reviewed translations and edited content as needed to ensure scientific accuracy, and they take full responsibility for the content of the publication.

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Appendix A. Supplementary data

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